

Queues

FIFO: FIRST IN, FIRST OUT

Linear Data Structures: Module 20

1. The Supermarket Logic

A **Queue** is a linear data structure that follows the **FIFO (First In First Out)** principle. It mirrors a real-life line at a shop: the first person to enter the line is the first one served.

FIRST IN → FIRST OUT



Queue Operations:

- **Enqueue:** Add an item to the Rear.
- **Dequeue:** Remove an item from the Front.
- **Peek:** Look at the Front item.
- **isEmpty:** Check if the line is empty.

2. Array Implementation

Using a Python list (array) is the simplest way to build a queue, though `pop(0)` has a performance cost as all other elements must shift left.

```
class Queue:
    def __init__(self):
        self.items = []

    def enqueue(self, item):
        self.items.append(item)

    def dequeue(self):
```

```
if not self.isEmpty():
    # In arrays, removing index 0 forces a shift
    return self.items.pop(0)
return "Queue is empty"

def isEmpty(self):
    return len(self.items) == 0
```

Warning: In large arrays, dequeue is slow ($O(n)$) because memory must be re-organized to fill the gap at index 0.

codeHive

3. Linked List Implementation

By using a Linked List with front and rear pointers, we achieve $O(1)$ performance for both enqueue and dequeue.

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

class Queue:
    def __init__(self):
        self.front = None
        self.rear = None

    def enqueue(self, data):
        new_node = Node(data)
        if self.rear is None:
            self.front = self.rear = new_node
            return
        self.rear.next = new_node
        self.rear = new_node

    def dequeue(self):
        if self.front is None: return "Empty"
        temp = self.front
        self.front = self.front.next
        if self.front is None: self.rear = None
        return temp.data
```

4. Where are Queues used?

Printer Spooling

Documents sent to a shared office printer are handled in a queue. The first file sent is the first file printed.

Breadth-First Search (BFS)

Queues are the engine behind BFS algorithms used in GPS navigation and social network "friend-of-friend" discovery.

5. Quick Quiz

Scenario: You have an array `[5, 11, 8, 3]` used as a queue. You call `enqueue(7)` then `dequeue()`. What value is removed?

Answer: 5 (The value at index 0 is always the first out).